In a real-world occlusion scenario (like a blind intersection with a large building or a parked truck blocking the view), a vehicle's onboard sensors have physical limitations. They cannot see through walls.

Here is the exact difference between how the car behaves in the **Unassisted** vs. **Assisted** configurations based on your project description:

**1. Unassisted Scenario (The Control Run)**

- **What the car does:** The ego vehicle drives entirely "blind" to the hidden danger. It completely ignores any MQTT/V2X messages being broadcast by the roadside camera.
- **Perception:** It relies *only* on its own onboard local camera. Because of the occlusion (the blind spot), its local camera cannot detect the pedestrian until the car clears the obstruction.
- **Behavior:** By the time the pedestrian enters the car's local line of sight, the vehicle is already too close (e.g., less than 14 meters away). The car is forced to slam on its emergency brakes at maximum force ($1.0$), resulting in a highly aggressive stop, a severe near-miss, or a full collision.

**2. Assisted Scenario (The V2X Run)**

- **What the car does:** The car actively listens to and utilizes the network.
- **Perception:** The virtual roadside camera is mounted high up on a pole or traffic light, giving it a clear bird's-eye view of the hidden side of the intersection. It detects the pedestrian early, packages that data into a compact MQTT/V2X message (with world coordinates and timestamps), and sends it out.
- **Behavior:** The vehicle receives this message while it is still far away (e.g., 35 meters out). Even though its own onboard camera still sees a blank wall, the car trusts the infrastructure data. It executes its "slow-down policy"—applying a smooth, controlled braking force ($0.5$) early on—and brings the vehicle to a safe, comfortable stop well before reaching the crosswalk.

**Why this matters for your Report**

When you run the script, the table compiles these exact differences into hard metrics to prove the value of V2X:

- **Warning Lead Time:** This measures how much earlier the V2X system caught the hazard. You will likely see numbers like $+1.5\text{s}$ or $+2.0\text{s}$ in the assisted column.
- **Speed Reduction:** In the unassisted column, the speed at brake trigger will be high (the car was cruising at full speed when it suddenly saw the person). In the

assisted column, the vehicle speed will already be smoothly decaying or completely stopped safely.

- **Safety Outcome:** The unassisted column will highlight the risk of COLLISION or a close NEAR-MISS, while the assisted column will consistently show AVOIDED.